# WORK SAMPLE — TRISHIR KULKARNI

Two mechanism projects: a built stair-climbing rover, and a cycloidal speed reducer design study

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## PROJECT 1 — STAIR-CLIMBING ROVER MECHANISM

Six-wheel rocker-bogie platform with spider-leg wheel arms — design, procurement, build and test. MGM's Polytechnic, Aurangabad, 2019–2020. Published: IRJET Vol. 07 Issue 06, June 2020, pp. 7422–7424.

**My role.** Four-student team with a faculty guide; I led it. I was responsible for the mechanism design and layout, sourcing and procuring every component, the physical assembly, and the writing of both the project report and the published paper. Electronics wiring and control setup were shared with a teammate.

### THE PROBLEM

Moving a load up a staircase normally means either carrying it or installing infrastructure. We set out to build a small remote-operated platform that could climb a domestic staircase and cross rough ground carrying a payload, using components a student team could actually buy and assemble — no machined parts, no custom castings, and a budget in the low tens of euros.

The design driver was **keeping all wheels loaded on an uneven surface**. A rigid six-wheel chassis loses traction the moment one wheel lifts, and a sprung suspension long enough to absorb a stair riser would make the vehicle unstable. The mechanism had to redistribute the body over the obstacle instead of absorbing it.

### CONCEPT SELECTION

We evaluated three approaches against climbable step height, stability under load, part availability, and build complexity:

- **Tracked platform** — good obstacle performance, but the track and tensioner were the whole project on their own and could not be sourced locally.
- **Rigid six-wheel chassis with sprung suspension** — simple to build, but wheel lift on a stair riser costs traction exactly when it is needed most.
- **Rocker-bogie linkage** — a passive, unsprung differential linkage that lets one wheel pair climb while the remaining wheels keep the body level and loaded. No actuator, no controller, no energy storage in the suspension.

We selected the rocker-bogie. The decisive argument was that it solves the traction problem *passively*: the geometry does the work, so nothing has to be sensed, controlled, or tuned. For a four-person student team on a fixed timeline, a mechanism with no control loop is worth more than a mechanism with better peak performance.

We then added four **spider-leg wheel arms** to the standard layout. The rocker-bogie alone is limited by wheel diameter relative to step height; extending the outer wheels on arms increases the effective reach over a riser without scaling up the whole vehicle.

### MECHANISM

The platform carries six driven wheels. On each side, a rocker arm and a bogie arm are joined by pivots, with a differential connection across the body so that the two sides can articulate independently while the body settles to the average of the two. When the leading wheel meets a riser, the remaining wheels drive forward and the linkage rotates the leading wheel up and over; the body pitch changes by roughly half of what a rigid chassis would experience.

Every wheel is independently driven, so the vehicle steers skid-style and there is no steering linkage to package. Drive is transmitted directly from a geared motor at each hub.

### STRUCTURE AND COMPONENT SELECTION

The frame is built from **PVC pipe and standard fittings**. This was a deliberate decision, not a compromise: the fittings give repeatable right-angle and elbow joints without any machining, the material is light enough that the 12 V drive motors are not working against the structure, and revising a link length during assembly meant cutting a new pipe section rather than re-fabricating a part. Cut lengths were standardised to six sizes so that the arms could be reconfigured while we iterated the geometry.

Component choices and the reasoning behind each:

- **Drive: 30 rpm 37 mm side-shaft DC gear motors, six off.** Selected for output torque at the wheel rather than speed — a stair climb is a low-speed, high-torque duty cycle, and the integral gearbox handles stall torque during a climb attempt. The 6 mm D-profile shaft couples directly to the wheel hub without an adapter.
- **Wheels: 125 mm rubber, rated to ~3 kg per four wheels.** Diameter was the constraint that mattered — the larger the wheel relative to the riser, the lower the climbing effort. Soft rubber gave the traction the linkage depends on.
- **Control: Arduino UNO R3-based dual-channel 20 A robot control board with a PS2 wireless receiver.** One board carrying microcontroller and motor drivers meant fewer interconnects to fail during assembly, and the 20 A per channel rating left margin for stall currents during a climb.
- **Power: 12 V sealed lead-acid battery.** Chosen for cost and current delivery; its mass sits low in the chassis, which helps the pitch stability the linkage is trying to preserve.



*Complete parts set before assembly: six geared drive motors, PS2 controller and receiver, the Arduino-based motor control board, cut PVC sections and fittings, and the drive wheels.*

## PROCUREMENT AND BUILD

I sourced the full bill of materials, which in practice meant reconciling what the design wanted with what was actually available locally and within budget. Two decisions came directly out of that: the wheel diameter was set by what could be bought off the shelf rather than by an ideal sizing calculation, and the motor count was fixed at six identical units so that a single spare covered every position on the vehicle.

Assembly was done by hand: cutting and dry-fitting the PVC structure, mounting the motors, wiring the drives back to the control board, and mounting the battery. We dry-fitted the entire linkage before bonding any joint, which let us adjust arm lengths against a real staircase rather than against a drawing.

## RESULTS

The completed vehicle climbed domestic stairs under remote control and crossed uneven ground and raised edges, with the linkage keeping the body substantially more level than a rigid chassis would. Skid steering worked as intended on both flat ground and loose surfaces.

**It did not meet its own target on tall steps.** On risers approaching and exceeding the wheel diameter, the leading wheel tended to push into the face of the step rather than climb it, and the vehicle stalled. This is the honest limit of the build and it is stated as such in the published paper.

## WHAT I WOULD DO DIFFERENTLY

This project was built at diploma level and I would approach several parts of it differently now:

- **Size the geometry before buying anything.** We selected wheels on availability and then discovered the step-height limit empirically. The relationship between wheel radius, arm length and maximum climbable riser is a straightforward

geometric and static analysis, and doing it first would have set the wheel specification instead of the other way round.

- **Analyse the stall case.** The failure on tall risers is a traction and approach-angle problem — the contact normal at the step face passes above the point where available friction can generate lift. A free-body analysis of the leading wheel at first contact would have predicted the limit before the build, and would have pointed at the fix.
- **Fix the structural compliance.** Bonded PVC joints flex under load, which costs some of the linkage travel the mechanism depends on. Aluminium extrusion or printed joints with a metal tube would hold the geometry without adding much mass.
- **Instrument it.** We assessed performance by observation. Current draw per motor during a climb would have identified which wheel was actually doing the work and whether the differential was sharing load as intended — a measurement that costs one shunt per channel.

## WHY THIS PROJECT IS RELEVANT TO HUMANOID HARDWARE

The transferable content is not the rover. It is that the mechanism was selected because it solved a problem passively rather than through control, that the structural concept was chosen for iteration speed during a build, and that the limitation was found by testing hardware rather than assumed away. In gripper and hand development the same trade-offs recur constantly: what to solve in geometry versus in control, how to build something that can be revised on the bench, and how quickly a design decision can be falsified by a real part.

It is also the project where I did the sourcing, the assembly and the writing — the unglamorous half of prototype work, and the half that determines whether anything reaches a testable state.

Published as: N. D. Bankar, D. Faiz, T. Kulkarni, Y. Kulkarni, S. Jagtap, “Design and Fabrication of Stair Climbing Mechanism”, IRJET Vol. 07 Issue 06, June 2020, pp. 7422–7424.

# PROJECT 2 — CYCLOIDAL SPEED REDUCER (DESIGN STUDY)

Two-stage cycloidal drive — concept, disc geometry, component design and analysis. B.E. dissertation carried out at DRDO R&DE (Engineers), Pune, 2022–2023, under Mr. Dipak Bhirud.

**My role.** Three-student dissertation team. I produced the SolidWorks CAD models and the cycloidal disc profile geometry, ran the finite-element analysis, and contributed substantially to the reduction-ratio calculations, the motion simulation and the writing of the report. **This was a design and analysis study — no hardware was built or tested**, so nothing below is presented as measured performance.

**Confidentiality.** The work was carried out at a DRDO establishment. The application, the design input values, the envelope constraints and the numerical results are withheld. What follows describes only the engineering method and my own contribution to it.

## WHY A CYCLOIDAL DRIVE

A cycloidal reducer achieves a high reduction ratio in one compact coaxial stage, using rolling contact between a lobed disc and a ring of housing rollers instead of meshing gear teeth. The properties that make it worth the geometric complexity are low backlash, tolerance of shock loading because load is shared across many contact points at once, and a short axial length relative to the ratio achieved. Those are the same reasons the architecture appears in robot joint modules, and the reason I wanted to work on it.

## APPROACH

The study followed a benchmark → concept → geometry → component design → analysis sequence:

- **Benchmarking.** We reviewed the published literature on cycloidal drive design and analysis and examined a commercially available cycloidal reducer family to establish what proportions and construction standard practice actually uses, rather than deriving everything from first principles in isolation.
- **Concept generation.** We developed and compared a single-stage and a multi-stage layout. The multi-stage arrangement was selected: splitting the overall reduction across two stages keeps the lobe count and disc diameter of each stage within sensible bounds, where forcing the whole ratio into one stage drives disc size and contact stress up sharply.
- **Disc geometry — my main contribution.** The cycloidal profile is generated from the rolling-circle diameter, the eccentricity and the roller pitch circle; these are not free parameters but a linked set, and changing one to suit packaging forces the others. I generated the profiles for both stage ratios in SolidWorks and sized the output-pin holes in the disc, which govern how the eccentric motion is converted to concentric output.
- **Component design.** Input shaft sized from its shear-force and bending-moment distribution, eccentric bearing selected for the rotating eccentric load, and the housing roller ring dimensioned against the disc profile.

## ANALYSIS

Four tools were used, each answering a different question:

- **MATLAB** — the distribution of contact force between the disc and the housing rollers, and the input-to-output rotation relationship. This established which rollers carry load at a given angular position and therefore where the design is actually stressed.
- **Recurdyn** — multibody dynamic simulation with explicit contact definitions between the moving bodies, to confirm the mechanism kinematics and transmit realistic loads to the structural model.
- **ANSYS** — static structural analysis of the cycloidal disc, casing, lid, input shaft and output shaft, evaluating stress and total deformation. This was my part of the work.
- **SolidWorks motion** — assembly-level motion checking and interference detection across the full rotation.

Because no hardware existed, validation could not be experimental. Instead we cross-checked the software results against independent theoretical hand calculations — agreement between the two was the acceptance criterion. We also worked through the drive's credible failure modes and checked the design against each.

## HONEST LIMITS OF THIS WORK

- It is a design study. Nothing was manufactured, so there is no measured backlash, no transmission error data, and no efficiency figure.

- Contact analysis was static. Cycloidal drives are sensitive to how load transfers between rollers during rotation, and a quasi-static sweep through the cycle would have characterised that better than discrete load cases.
- Manufacturing tolerance was not carried through the analysis. Eccentricity is the parameter the whole geometry hangs on, and the achievable tolerance on it determines real backlash — a tolerance stack-up on that dimension is the first thing the study needed and did not have.

## WHAT I WOULD DO NEXT

Build a single stage and measure it. The two numbers that decide whether a cycloidal joint drive is usable — backlash and transmission error under reversing load — are exactly the ones no amount of simulation settles convincingly. A simple rig with an encoder on each shaft and a reversing load would produce them, and would show whether the eccentricity tolerance assumption held. I would also run a proper mesh convergence study on the disc contact region and add a thermal and lubrication assessment, neither of which the study covered.

## RELEVANCE TO HUMANOID HARDWARE

A robot finger or wrist joint needs high reduction in a very small envelope with backlash tight enough that position control means something. That is the cycloidal problem in miniature, with mass and packaging constraints that are harder still. This project is where I learned how the geometry, the contact loading and the achievable tolerance are tied together in a joint drive — and, from the rover project, what happens to a design that has not been checked against real hardware. Those two lessons are why I want to work on gripper and hand development.